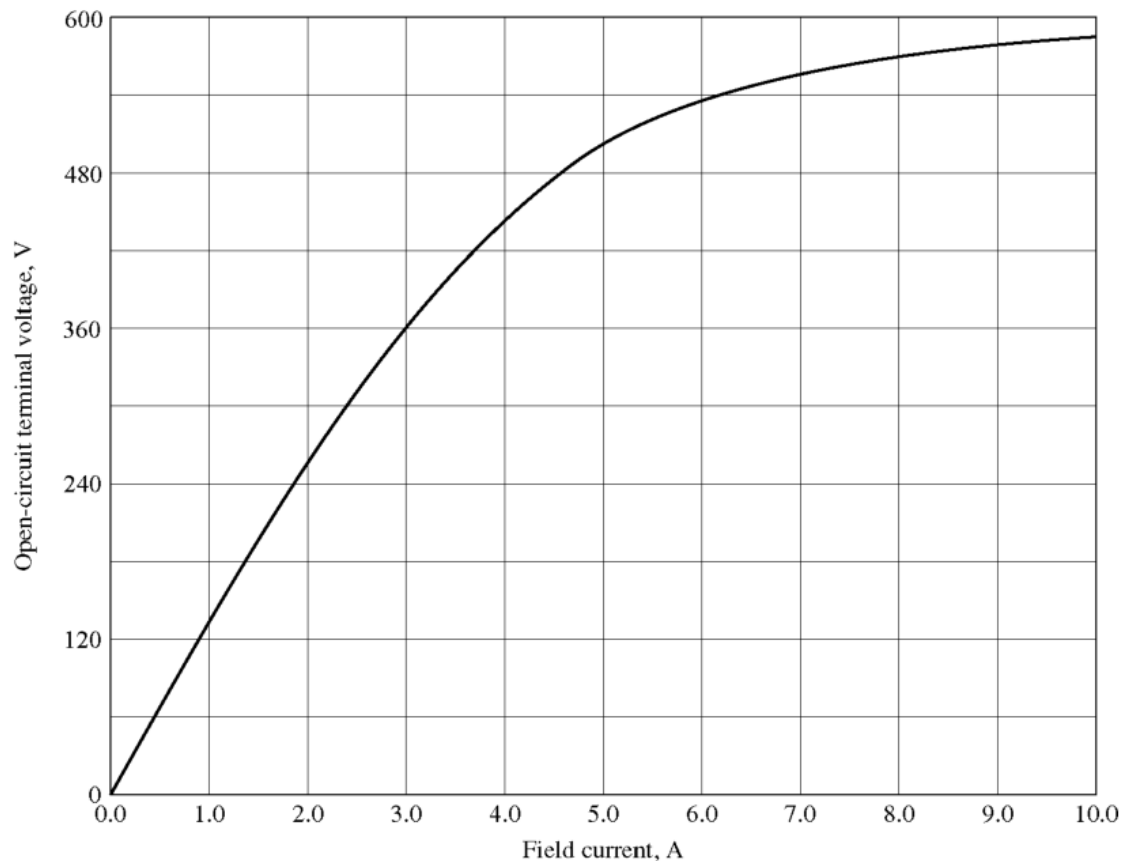
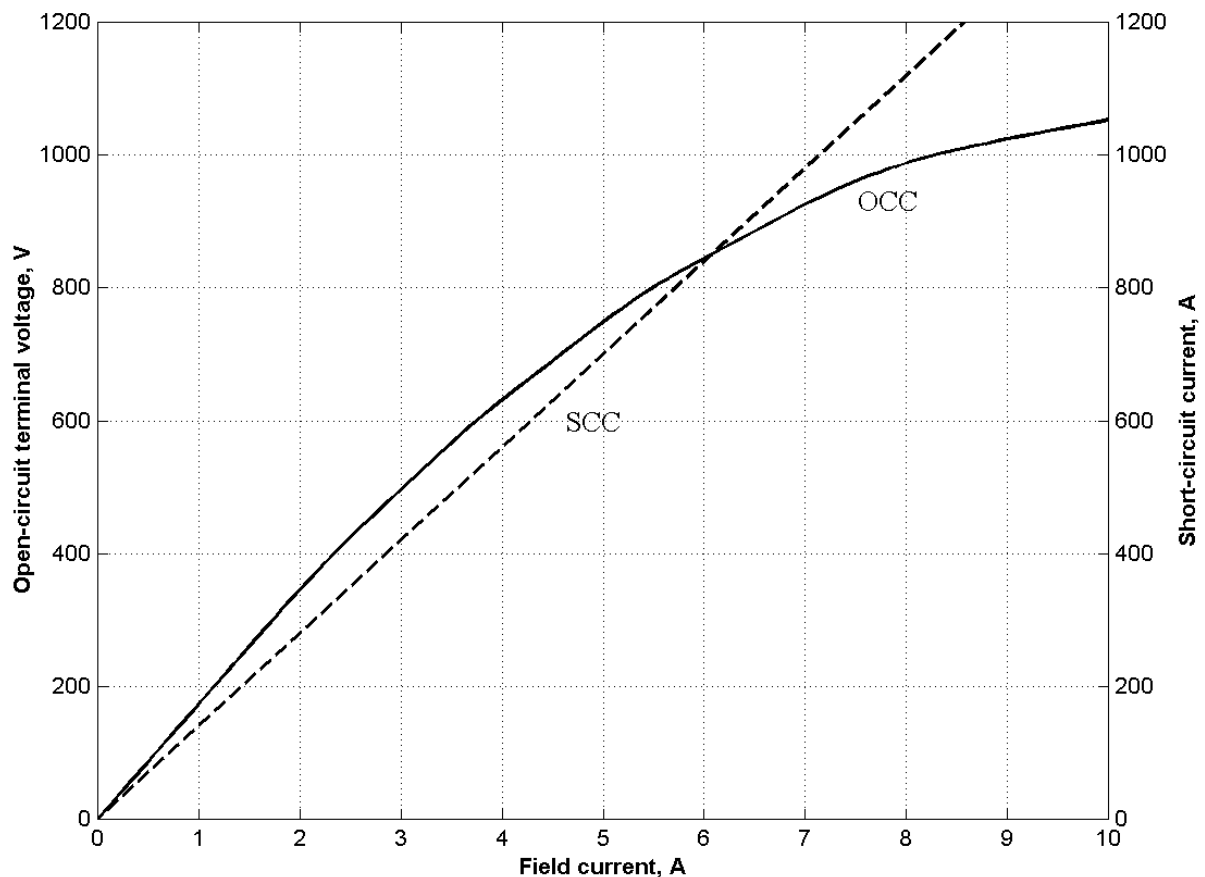


Exercises 6 – Synchronous Generator

1. A 480-V 200-kVA 0.8-power-factor-lagging 60-Hz two-pole Y-connected synchronous generator has a synchronous reactance of $0.25\ \Omega$ and an armature resistance of $0.03\ \Omega$. At 60 Hz, its friction and windage losses are 6 kW, and its core losses are 4 kW. The field circuit has a dc voltage of 200 V, and the maximum I_F is 10 A. The resistance of the field circuit is adjustable over the range from 20 to 200 Ω . The OCC of this generator is shown in figure below.
 - a. How much field current is required to make V_T equal to 480 V when the generator is running at no load?
 - b. What is the internal generated voltage of this machine at rated conditions?
 - c. How much field current is required to make V_T equal to 480 V when the generator is running at rated conditions?
 - d. How much power and torque must the generator's prime mover be capable of supplying?



2. A two-pole Y-connected synchronous generator rated at 300 kVA, 480 V, 60 Hz, and 0.85 PF lagging. Its armature resistance R_A is $0.04\ \Omega$ and synchronous reactance is $0.695\ \Omega$. The core losses of this generator at rated conditions are 10 kW, and the friction and windage losses are 13 kW. The open-circuit and short-circuit characteristics are shown in figure below.



- What are the rated current and internal generated voltage of this generator?
 - What field current does this generator require to operate at the rated voltage, current, and power factor?
 - What is the voltage regulation of this generator at the rated current and power factor?
 - If this generator is operating at the rated conditions and the load is suddenly removed, what will the terminal voltage be?
 - What are the electrical losses in this generator at rated conditions?
 - If this machine is operating at rated conditions, what input torque must be applied to the shaft of this generator?
 - Find the efficiency of the generator?
3. A 480-V 200-kW two-pole three-phase 50-Hz synchronous generator's prime mover has a no-load speed of 3040 r/min and a full-load speed of 2975 r/min. It is operating in parallel with a 480-V 150-kW four-pole 50-Hz synchronous generator whose prime mover has a no-load speed of 1500 r/min and a full-load speed of 1485 r/min. The loads supplied by the two generators consist of 200 kW at 0.85 PF lagging.
- Calculate the speed droops of generator 1 and generator 2.
 - Find the operating frequency of the power system.
 - Find the power being supplied by each of the generators in this system.